

BOND PARAMETERS

JEE Advanced Preparation - Chemistry

Bond Length, Bond Angle, Bond Enthalpy, Bond Order, Dipole Moment

Topic Overview

Bond parameters describe the measurable properties of chemical bonds. Understanding bond length, bond angle, bond enthalpy, bond order, and dipole moment is essential for predicting molecular stability, reactivity, and polarity in JEE Advanced.

1. BOND LENGTH

Bond length is the **equilibrium distance between the nuclei of two bonded atoms**. It is measured in picometres (pm) or Angstroms (1 Å = 100 pm).

Factors affecting Bond Length:

- Bond Order: Higher bond order → shorter bond length (more electrons shared, nuclei pulled closer)
- Atomic Size: Larger atoms → longer bonds (e.g., C-I > C-Br > C-Cl > C-F)
- Hybridisation: More s-character → shorter bond (sp > sp² > sp³ character → shorter bond)
- Resonance: Resonance structures have bond lengths intermediate between single and double

2. BOND ENTHALPY (BOND ENERGY)

Bond enthalpy is the **energy required to break one mole of a particular bond** in gaseous molecules into gaseous atoms. It is always a positive value (endothermic process). Also called Bond Dissociation Energy.

Key Relationships:

- Higher bond order → higher bond enthalpy (more energy needed to break)
- Shorter bond → higher bond enthalpy
- For polyatomic molecules, average bond enthalpy is used
- enthalpy of reaction = sum of bonds broken - sum of bonds formed

3. BOND ORDER

Bond Order = (Bonding electrons - Antibonding electrons) / 2
Single bond: BO = 1 | Double bond: BO = 2 | Triple bond: BO = 3
Higher BO → shorter length → higher enthalpy → stronger bond

4. COMPARATIVE DATA TABLE

Bond	Bond Order	Length (pm)	Enthalpy (kJ/mol)
C-C	1	154	347
C=C	2	134	611
C≡C	3	120	837
N-N	1	145	163

Bond	Bond Order	Length (pm)	Enthalpy (kJ/mol)
N=N	2	125	418
N≡N	3	110	945
O-O	1	148	146
O=O	2	121	498
C-H	1	109	414
O-H	1	96	460
N-H	1	101	391
C-F	1	135	485
C-Cl	1	177	327
C-Br	1	194	285
H-H	1	74	436
H-F	1	92	569
H-Cl	1	127	432

5. RESONANCE AND BOND PARAMETERS

Resonance Effect on Bond Lengths:

- O₃ (ozone): O-O bond length = 128 pm — between O-O (148 pm) and O=O (121 pm)
- Benzene: All C-C bonds equal 140 pm — between C-C (154 pm) and C=C (134 pm)
- CO₃²⁻: All C-O bonds equal ~129 pm — between C-O (143 pm) and C=O (120 pm)
- Resonance stabilises molecules and equalises bond lengths

6. DIPOLE MOMENT

Definition: $\mu = q \times d$ (Debye)

- μ = dipole moment (unit: Debye, D; $1 \text{ D} = 3.336 \times 10^{-30} \text{ C.m}$)
- q = magnitude of partial charge (in coulombs)
- d = bond length / distance between charges (in metres)

Direction (Chemistry convention): Vector points from positive (+) end to negative (-) end, i.e., from less electronegative to more electronegative atom.

Physics convention: Opposite direction (negative to positive). JEE follows Chemistry convention.

6.1 Zero Dipole Moment Molecules

Zero Dipole Moment (symmetric molecules):

- BeCl_2 : Linear (180°) — two equal bond dipoles cancel
- BF_3 : Trigonal planar (120°) — three equal bond dipoles cancel
- CCl_4 : Tetrahedral (109.5°) — four equal bond dipoles cancel
- CO_2 : Linear (180°) — two C=O dipoles cancel
- SF_6 : Octahedral (90°) — six S-F dipoles cancel
- PF_5 : Trigonal bipyramidal — all bond dipoles cancel
- trans-CHCl=CHCl : Bond dipoles cancel by symmetry

6.2 Non-Zero Dipole Moment Molecules

Non-Zero Dipole Moment (asymmetric / lone pairs):

- HCl : 1.08 D — H-Cl single bond, Cl more electronegative
- HF : 1.91 D — H-F bond, F is most electronegative element
- H_2O : 1.84 D — bent shape, lone pairs add to net dipole

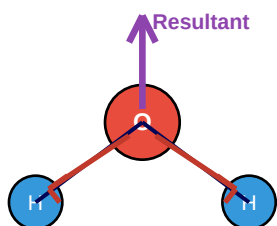
- NH_3 : 1.46 D — trigonal pyramidal, lone pair contributes
- CHCl_3 : 1.01 D — three C-Cl dipoles add to give non-zero net
- CH_3Cl : 1.87 D — only one C-Cl bond
- cis-CHCl=CHCl : Non-zero (dipoles on same side add up)

6.3 Electronegativity and Bond Polarity

Bond polarity depends on the **electronegativity difference** between atoms. The greater the difference, the more polar the bond. Pauling scale: $\text{F}(4.0) > \text{O}(3.5) > \text{N}(3.0) > \text{Cl}(3.0) > \text{Br}(2.8) > \text{C}(2.5) > \text{H}(2.1) > \text{Si}(1.8)$

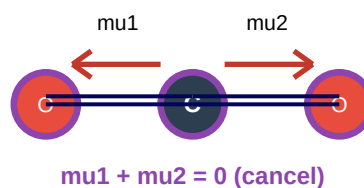
7. DIPOLE MOMENT DIAGRAMS

H_2O — Net Dipole NOT zero



Net dipole NON-ZERO (1.84 D)

CO_2 — Net Dipole = zero



Net dipole = 0 D (linear, symmetric)

8. WORKED EXAMPLES

Example 1 — Compare bond lengths in N₂, N₂H₄, N₂H₂

Question: Arrange N-N, N=N, N≡N in increasing bond length.

Solution:

- N-N single bond (hydrazine N₂H₄): bond order 1, length = 145 pm
- N=N double bond (diazene N₂H₂): bond order 2, length = 125 pm
- N≡N triple bond (nitrogen N₂): bond order 3, length = 110 pm
- Increasing bond order → decreasing bond length
- Answer: N≡N < N=N < N-N (110 < 125 < 145 pm)

Example 2 — Bond order and length in O₃ vs O₂

Question: Compare the O-O bond length in O₃ (ozone) and O₂ (oxygen).

Solution:

- O₂: double bond (O=O), bond order = 2, bond length = 121 pm
- O₃: resonance hybrid of two structures with one O-O and one O=O
- Fractional bond order in O₃ = 1.5 (average of 1 and 2)
- Bond length in O₃ = 128 pm (between 148 pm for O-O and 121 pm for O=O)
- Answer: O-O bond in O₃ (128 pm) is LONGER than in O₂ (121 pm)

Example 3 — Dipole moment of CO₂ vs SO₂

Question: Why is the dipole moment of CO₂ zero while SO₂ is non-zero?

Solution:

- CO₂: Linear (sp, 180°). Two C=O bond dipoles are equal and opposite.
- They exactly cancel → net dipole = 0 D
- SO₂: Bent/V-shaped (sp², ~119°). Two S=O bond dipoles do NOT cancel.
- The molecule has a net dipole pointing away from S toward the two O atoms.
- Additionally, S has a lone pair which contributes to the overall asymmetry.
- Answer: CO₂ is linear (symmetric) → zero dipole; SO₂ is bent (asymmetric) → non-zero dipole (~1.63 D)

Example 4 — Comparing bond enthalpies

Question: Arrange C-C, C=C, C≡C in decreasing order of bond enthalpy.

Solution:

- C-C: bond order 1, enthalpy = 347 kJ/mol
- C=C: bond order 2, enthalpy = 611 kJ/mol
- C≡C: bond order 3, enthalpy = 837 kJ/mol
- Higher bond order → more electrons shared → stronger bond → higher enthalpy
- Answer: C≡C > C=C > C-C (837 > 611 > 347 kJ/mol)

Example 5 — Predict whether CHCl_3 has a dipole moment

Question: Does chloroform (CHCl_3) have a non-zero dipole moment?

Solution:

- Structure: 3 C-Cl bonds and 1 C-H bond in tetrahedral arrangement
- C-Cl bond: Cl is more electronegative $\rightarrow$ bond dipole points toward Cl
- C-H bond: H is less electronegative $\rightarrow$ bond dipole points toward C
- In CCl_4 : 4 equal C-Cl dipoles cancel $\rightarrow$ zero dipole
- In CHCl_3 : 3 C-Cl dipoles do not fully cancel (the C-H replaces one C-Cl)
- The three C-Cl dipoles combine to give a resultant that does NOT cancel the C-H dipole
- Answer: CHCl_3 has a non-zero dipole moment ($\sim 1.01 \text{ D}$)

9. KEY FACTS — QUICK REVISION

KEY FACTS / QUICK REVISION — BOND PARAMETERS

- Bond length order: single > double > triple (same atoms)
- Bond enthalpy order: single < double < triple (same atoms)
- Higher bond order → shorter length, higher enthalpy, stronger bond
- Benzene C-C: 140 pm (resonance between 134 and 154 pm)
- O₃ O-O: 128 pm (resonance between 121 and 148 pm)
- Dipole moment $\mu = q \times d$; unit = Debye (D)
- Zero dipole: BeCl₂, BF₃, CCl₄, CO₂, SF₆ (all symmetric)
- Non-zero dipole: HF (1.91D), H₂O (1.84D), HCl (1.08D), NH₃ (1.46D)
- cis-isomers have higher dipole than trans-isomers
- Lone pairs on central atom contribute to net dipole (e.g., NH₃, H₂O)
- Electronegativity: F > O > N = Cl > Br > C > H (Pauling scale)

10. PRACTICE QUESTIONS — JEE ADVANCED LEVEL

1. The correct order of C-C bond length is:

- (A) Benzene > Ethane > Ethylene
- (B) Ethane > Benzene > Ethylene
- (C) Ethylene > Benzene > Ethane
- (D) Ethane > Ethylene > Benzene

2. Which of the following has the highest bond enthalpy?

- (A) N-N
- (B) N=N
- (C) N≡N
- (D) O=O

3. The dipole moment of CO₂ is zero because:

- (A) C is in the centre
- (B) It is a linear molecule with equal and opposite dipoles
- (C) Electronegativity of C and O are equal
- (D) CO₂ has no polar bonds

4. Which of the following has a non-zero dipole moment?

- (A) BF₃
- (B) CCl₄
- (C) NF₃
- (D) CO₂

5. Bond length in ozone (O₃) compared to O₂ is:

- (A) Shorter
- (B) Longer
- (C) Equal
- (D) Cannot be compared

6. The bond order in benzene is:

- (A) 1
- (B) 1.5
- (C) 2
- (D) 2.5

7. Which factor does NOT affect bond length?

- (A) Bond order
- (B) Atomic size
- (C) Temperature at STP
- (D) Hybridisation

8. The correct order of bond enthalpy is:

- (A) C-F < C-Cl < C-Br
- (B) C-F > C-Cl > C-Br
- (C) C-Cl > C-F > C-Br
- (D) All are equal

9. Dipole moment of cis-2-butene compared to trans-2-butene:

- (A) Equal
- (B) cis > trans
- (C) trans > cis
- (D) Both zero

10. The bond angle in water is 104.5° . This is primarily due to:

- (A) Small size of H
- (B) Two lone pairs on O causing lp-bp repulsion
- (C) sp hybridisation of O
- (D) Weak O-H bonds

11. ANSWER KEY WITH EXPLANATIONS

1. Answer: B — Ethane > Benzene > Ethylene

Ethane (C-C single): 154 pm. Benzene (resonance): 140 pm. Ethylene (C=C double): 134 pm. Higher bond order = shorter bond.

2. Answer: C — N≡N

N≡N has triple bond (bond order 3) → highest bond enthalpy = 945 kJ/mol. O=O = 498 kJ/mol; N=N = 418 kJ/mol; N-N = 163 kJ/mol.

3. Answer: B

CO₂ is linear (sp hybridised). The two C=O bond dipoles are equal in magnitude but opposite in direction, so they exactly cancel giving net dipole = 0.

4. Answer: C — NF₃

NF₃ has pyramidal shape with lone pair on N. The three N-F bond dipoles do not cancel, and lone pair contribution makes dipole non-zero (~0.24 D). BF₃, CCl₄, CO₂ all have zero dipole.

5. Answer: B — Longer

O₃ has resonance with fractional bond order 1.5 (between 1 and 2). Bond length = 128 pm > O=O bond length in O₂ (121 pm).

6. Answer: B — 1.5

Benzene has 3 sigma bonds and 3 pi bonds delocalised over 6 C-C bonds. Average bond order = (3+3)/6 ... actually each C-C has order 1.5 (intermediate between single and double).

7. Answer: C — Temperature at STP

Bond length is a property of the bond itself (ground state equilibrium distance). It is determined by bond order, atomic size, and hybridisation — not by temperature under standard conditions.

8. Answer: B — C-F > C-Cl > C-Br

C-F: 485 kJ/mol; C-Cl: 327 kJ/mol; C-Br: 285 kJ/mol. Shorter bonds (smaller halogen) have higher bond enthalpy. F is smallest → C-F is shortest and strongest.

9. Answer: B — cis > trans

In cis-2-butene, the two CH₃ groups are on the same side — their dipoles add up. In trans-2-butene, they are on opposite sides and partially cancel. So cis has higher dipole moment.

10. Answer: B

O in H₂O has 2 lone pairs (AX₂E₂). Lone pair–bond pair repulsion (lp-bp) is stronger than bp-bp repulsion, compressing the H-O-H angle from 109.5° to 104.5°.